

Table of contents

Overview – page 1

v2.0.0 updates based on OrthoFinder analysis with new Asparagaceae genomes – page 1

OrthoFinder samples/references – page 2

Specific loci updates – page 2-3

HybPiper results compared between v1.1 and v2.0.0 – page 4

Overview

File contents include methods used to update the Asparagaceae1726 DNA coding sequence (CDS) target reference sequence (fasta) file used for read mapping (e.g., with HybPiper). This version decreases the number of paralogs recovered in Asparagaceae1726 datasets and increases on-target ortholog recovery *in silico* (i.e., with no changes to the original probe design). Below are methods regarding tests performed to increase ortholog representation and decrease paralog recovery for Asparagaceae1726 targets.

This target file version is 2.0.0.

The target file used in Bentz et al. (2024) (<https://doi.org/10.1093/gbe/evae200>) and Bentz and Leebens-Mack (2024) (<https://doi.org/10.1002/aps3.11597>) was version 1.1. Target loci in v2.0.0 were renamed numerically (i.e., “0001...1726”). Original loci names from v1.1 and their newly assigned names can be found in the file ‘target_loci_names.txt’.

v2.0.0 updates based on OrthoFinder analysis with new Asparagaceae genomes

Newly available Asparagaceae reference genomes (see list below) were used to reassess ortholog copy number variation and paralogy using OrthoFinder v2.5.4 with the MSA (multiple sequence alignment) option (<https://github.com/davidemms/OrthoFinder>).

Filters and updates:

- a) Targets were removed if >1 copy was detected in >70% of Asparagaceae samples
- b) Reference sequences from *Yucca aloifolia* and *Asparagus officinalis* (from v1.1) were replaced with updated annotations from more recent genome annotations (see list below)
- c) All targets lacking a start codon were removed
- d) All targets with an internal stop codon were removed
- e) Targets <20% of the average length compared to other orthologs were removed (only when multiple orthologs were present in target file; performed via HybPiper fix_targetfile)
- f) Complex targets (minimum threshold of 1.25) were removed according to HybPiper fix_targetfile (only when multiple orthologs were present in target file)
- g) Targets were removed if assigned to disjunct Orthogroups, relative to the original probe reference sequence (from either *A. setaceus* or *A. officinalis*); of which were considered to be paralogs. These putative paralogs were either assigned to incorrect Orthogroups in previous analyses, due to less representation of Asparagaceae genomes, or may represent erroneous gene predictions
- h) A proportion of putative paralogs were reincorporated into the target sequence fasta file as new/alternative targets, when supported by >2 Asparagaceae samples

OrthoFinder samples/references

1. *Dracaena cambodiana* (<https://doi.org/10.1038/s41597-024-03670-w>)
isolate: NGDC Genome Sequence Archive=GWHEQIC000000000 (project=PRJCA021348)
<https://ngdc.cncb.ac.cn/>
2. *Agave tequilana* (https://phytozome-next.jgi.doe.gov/info/Atequilanavar_WebersBlueHAP1_v2_1)
isolate: Atequilanavar_WebersBlueHAP1_790_v2.1 (DOE-JGI)
3. *Yucca filamentosa* (https://phytozome.jgi.doe.gov/info/YfilamentosaC3pri_v2_1)
isolate: YfilamentosaC3pri_837_v2.1 (DOE-JGI)
4. *Ananas comosus* (https://phytozome-next.jgi.doe.gov/info/Acomosus_v3)
(<https://doi.org/10.1038/ng.3435>)
isolate: Acomosus_321_v3
 - Used as non-Asparagaceae outgroup
5. *Asparagus setaceus* (<https://doi.org/10.5061/dryad.1c59zw3rm>).
<https://doi.org/10.1038/s41438-020-0271-y>
6. *Yucca aloifolia* (https://phytozome.jgi.doe.gov/info/YaloifoliaYa24Inoko_v2_1)
isolate: YaloifoliaYa24Inoko_839_v2.1
7. *Asparagus horridus* (Bentz et al., unpublished)
isolate: Aoff_pb32m_HAP2_v2.2
8. *Asparagus officinalis* (Bentz et al., unpublished)
isolate: Aoff_pb81m_HAP1_v1.0
9. *Dioscorea alata* (https://phytozome-next.jgi.doe.gov/info/Dalata_v2_1)
isolate: v2.1
 - Used as non-Asparagaceae outgroup
10. Asparagaceae1726_AA_w_orthologs.fa (v1.1)
(<https://github.com/bentzpc/Asparagaceae1726>)

Specific loci updates

v1.1 Targets lacking start codon:

Aofficinalis-OG0011577
Aofficinalis-OG0013948
Aofficinalis-OG0005955

v1.1 Targets with internal stop codon:

Asetaceus-OG0009300
Aofficinalis-OG0013948
Aofficinalis-OG0005955

v1.1 Targets with paralogs that were removed entirely:

Aofficinalis-OG0006683
Aofficinalis-OG0010852
Aofficinalis-OG0010956
Aofficinalis-OG0011453
Aofficinalis-OG0013122
Aofficinalis-OG0015797
Aofficinalis-OG0019078
Yaloifolia-OG0011582
Yaloifolia-OG0011605
Yaloifolia-OG0011729
Yaloifolia-OG0011932

Yaloifolia-OG0012226
Aofficinalis-OG0018994
Yaloifolia-OG0018994
Yaloifolia-OG0017479
Aofficinalis-OG0017479
Yaloifolia-OG0021458
Yaloifolia-OG0019173
Aofficinalis-OG0021482

v1.1 Targets with paralogs (that were reincorporated into v2.0.0 as new targets):

Asetaceus-OG0008945
Asetaceus-OG0010913
Asetaceus-OG0012276
Asetaceus-OG0012298
Asetaceus-OG0013349
Asetaceus-OG0014678
Asetaceus-OG0015797
Yaloifolia-OG0013537
Aofficinalis-OG0008203

v1.1 Multi-copy targets (>1 copy in 70% of Asparagaceae samples):

01_48
04_1258
04_460
06_374
07_742
07_1094 (07_1095 duplicate)
07_1095 (07_1094 duplicate)
OG0002673
OG0004252
OG0005103
OG0006814
OG0006929
OG0007287
OG0007572
OG0007735
OG0009486
OG0009799
OG0009828
OG0010366
OG0010814
OG0011140
OG0011436
OG0012775
OG0019079

***Note:** All low complexity targets were removed according to hybpiper check_targetfile, except for target **0180**, which would have been removed from the target list entirely.

HybPiper results compared between v1.1 and v2.0.0 (DNA CDS target references)

HybPiper results summary compared between Asparagaceae target files v1.1 and v2.0.0

Version	Species/sample	Total reads	Reads mapped	% on target	Genes recovered	Paralogs (long warning)	Paralogs (depth warning)
v1.1	<i>Asparagus officinalis</i> (PB_81)	14,242,278	4,641,539	32.6	1717	9	18
v1.1	<i>Aspidistra elatior</i> (PB_130)	38,730,368	3,610,995	9.3	1521	32	81
v1.1	<i>Behnia reticulata</i> (Burrows_15952)	16,600,390	2,398,381	14.4	1576	185	418
v1.1	<i>Drimiopsis botryoides</i> (CCH_154)	32,256,226	2,695,687	8.4	1416	7	20
v1.1	<i>Liriope muscari</i> (PB_129)	24,566,332	4,438,751	18.1	1596	50	157
v1.1	<i>Thysanotus</i> sp. (Thiele sn)	42,000,920	10,297,613	24.5	1468	47	145
v1.1	<i>Tritelia laxa</i> (Halse_8737)	39,880,246	5,567,362	14	1501	16	42
v1.1	<i>Yucca filamentosa</i> YF81	49,100,336	13,145,437	26.8	1698	166	467
v2.1	<i>Asparagus officinalis</i> (PB_81)	14,242,278	4,864,119	34.2	1699	5	15
v2.1	<i>Aspidistra elatior</i> (PB_130)	38,730,368	4,941,310	12.8	1548	26	93
v2.1	<i>Behnia reticulata</i> (Burrows_15952)	16,600,390	2,924,666	17.6	1571	144	388
v2.1	<i>Drimiopsis botryoides</i> (CCH_154)	32,256,226	3,585,509	11.1	1432	6	18
v2.1	<i>Liriope muscari</i> (PB_129)	24,566,332	5,340,606	21.7	1615	44	181
v2.1	<i>Thysanotus</i> sp. (Thiele sn)	42,000,920	11,730,118	27.9	1489	43	151
v2.1	<i>Tritelia laxa</i> (Halse_8737)	39,880,246	7,195,995	18	1505	17	45
v2.1	<i>Yucca filamentosa</i> YF81	49,100,336	14,486,571	29.5	1682	109	445

in silico recovery differences using v2.0.0 target file

Species/sample	% on target	Genes recovered	Paralogs (long warning)	Paralogs (depth warning)
<i>Asparagus officinalis</i> (PB_81)	+1.6	-18	-4	-3
<i>Aspidistra elatior</i> (PB_130)	+3.5	+27	-6	+12
<i>Behnia reticulata</i> (Burrows_15952)	+3.2	-5	-41	-30
<i>Drimiopsis botryoides</i> (CCH_154)	+2.7	+16	-1	-2
<i>Liriope muscari</i> (PB_129)	+3.6	+19	-6	+24
<i>Thysanotus</i> sp. (Thiele sn)	+3.4	+21	-4	+6
<i>Tritelia laxa</i> (Halse_8737)	+4	+4	+1	+3
<i>Yucca filamentosa</i> YF81	+2.7	-16	-57	-22